SYNTHETIC DATA — 교육용 가상 데이터이며 실제 제품 사양·운영 기준과 무관합니다.

Oil-immersed Transformer 1000kVA (TRF-OIL-1000) — Datasheet

1. Overview

Oil-immersed transformer 1000kVA, 11/0.433kV, ONAN, conservator type. Product family: Power transformers. Lifecycle status: **ACTIVE**.

2. Technical specifications

2.1. Ratings

Item	Value
Rated power	1000 kVA (ONAN)
Voltage ratio	11 kV / 0.433 kV
Vector group	Dyn11

2.2. Performance and characteristics

Item	Value
Impedance voltage	5% at 75 °C
Off-circuit tap range	±2 x 2.5%
Insulating liquid	Mineral oil (IEC 60296)

2.3. Endurance and communication

Item	Value
No-load loss	1,100 W
Load loss at 75 °C	9,500 W

2.5. Monitoring and auxiliaries

Item	Value
Top-oil temperature alarm / trip	90 °C / 100 °C
Protection devices	Buchholz relay, pressure relief device, oil & winding temperature indicators

3. Installation data

3.1. Terminals

- Dimensions (W x H x D): 1900 x 1300 x 2000 mm

- Weight: 3200 kg
- Degree of protection: IP54 (outdoor)
- Operating temperature: -25 to +40 °C

4. Standards and certifications

Applicable standards:

- IEC 60076-1
- IEC 60076-2

Certifications held:

- Type test report (IEC 60076)
- KC

Certificates and declarations of conformity are available from the Certification & Compliance team on request. Certifications not listed above are not held for this product; project-specific evaluation can be requested.

5. Ordering information

Item	Value
Ordering code	TRF-OIL-1000
Standard lead time	12 weeks ex-works
Minimum order quantity	1
Warranty	24 months (see POL-WARRANTY)
Firmware	not applicable

6. Related documents

- IG-TRF-OIL-1000 — Installation and Commissioning Guide
- TG-TRF-OIL-1000 — Troubleshooting and Error Code Guide
- MG-TRF-OIL-1000 — Maintenance Guide
- FAQ-TRANSFORMER — Power transformers FAQ

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Oil-immersed Transformer 1000kVA (TRF-OIL-1000) — Installation and Commissioning Guide

1. Safety

Installation, wiring and commissioning must be performed by qualified personnel. Disconnect and lock out all power sources before work. Verify absence of voltage.

2. Receiving and storage

Inspect the packaging on arrival and report damage within 7 days (see POL-RETURN). Store indoors in a dry place between -25 and +55 °C, relative humidity below 85% without condensation.

3. Mounting and clearances

- Required clearances: Minimum 500 mm from enclosure walls; maintain electrical clearances per IEC 61936-1
- Operating temperature: -25 to +40 °C
- Degree of protection: IP54 (outdoor)
- Above 1,000 m altitude, apply derating per the datasheet.

4. Wiring and tightening torques

- Terminal tightening torque: see dimension drawing
- Use copper conductors rated for at least 75 °C.
- Separate power and signal cables by at least 200 mm.

5. Commissioning procedure

1. Measure insulation resistance and polarisation index
2. Check off-circuit tap position against required secondary voltage
3. Verify temperature monitoring alarm and trip contacts
4. Check earthing of core and enclosure
5. Energise at no load and record secondary voltages

6. Pre-energisation checklist

☐ Product rating matches the drawing
☐ All terminals tightened to specified torque
☐ Protective earth connected
☐ Insulation resistance measured and recorded
☐ Settings / parameters recorded

7. Common commissioning issues

Symptom	Likely cause	See
Abnormal humming noise	Loose core clamping or harmonic load	TG-TRF-OIL-1000 §3.1
Buchholz alarm	Air trapped after oil filling	TG-TRF-OIL-1000 §3.2
Oil leakage at gasket	Gasket damage during transport	TG-TRF-OIL-1000 §3.4
Secondary voltage too high	Off-circuit tap at wrong position	TG-TRF-OIL-1000 §3.5

On-site commissioning support can be requested through the Global CS Center. For power transformers the request is handled by Field Service.

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Oil-immersed Transformer 1000kVA (TRF-OIL-1000) — Maintenance Guide

1. Maintenance intervals

Task	Interval
Visual inspection and cleaning	12 months
Insulation resistance	3 years
Temperature monitor function test	12 months
Oil DGA (oil-immersed)	12 months
Tightness of connections	12 months

2. Spare parts

- Gasket set

3. End of life

Lifecycle status: ACTIVE. Expected service life is 20 years under normal conditions. Dispose of according to local regulations for electrical equipment.